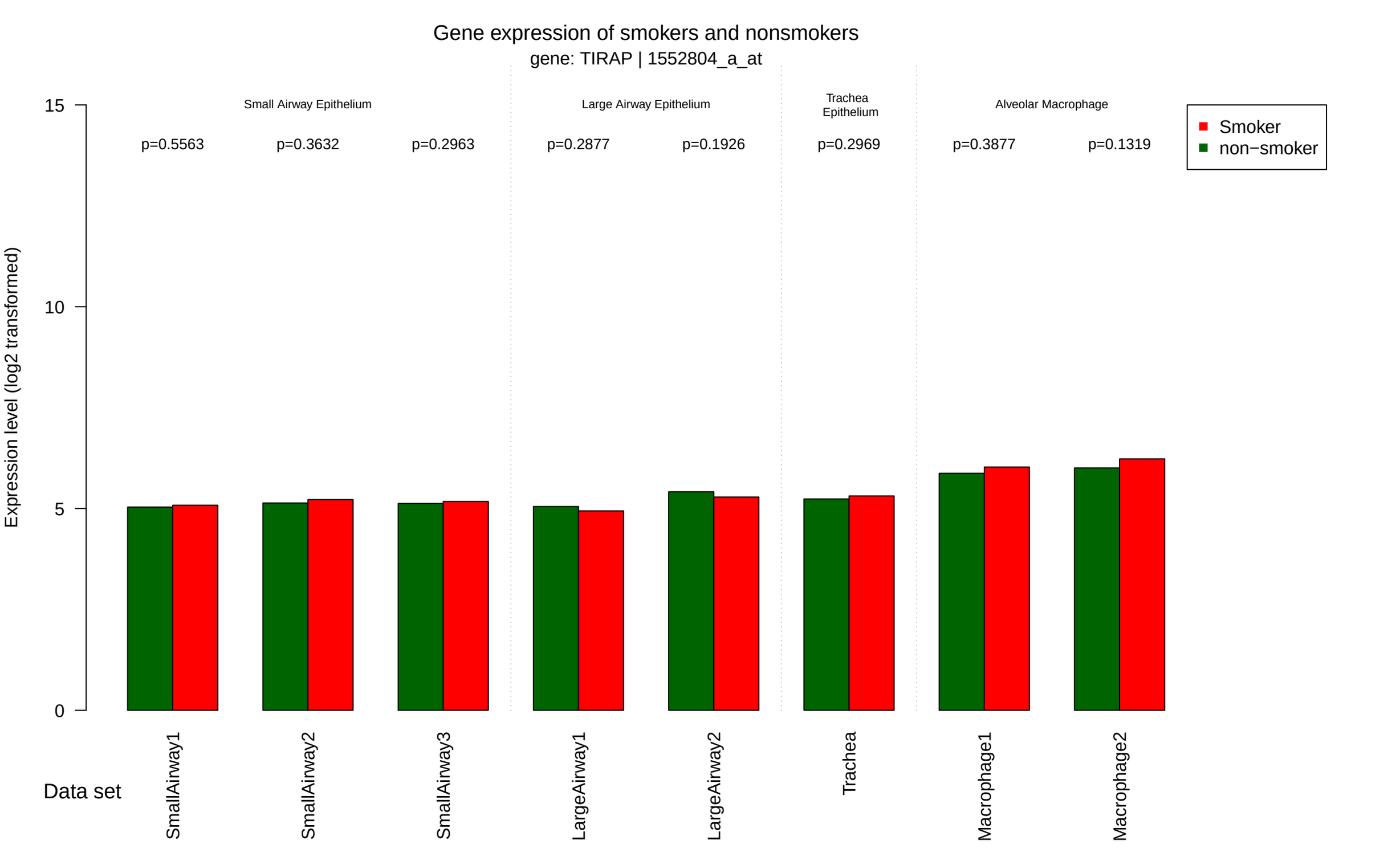
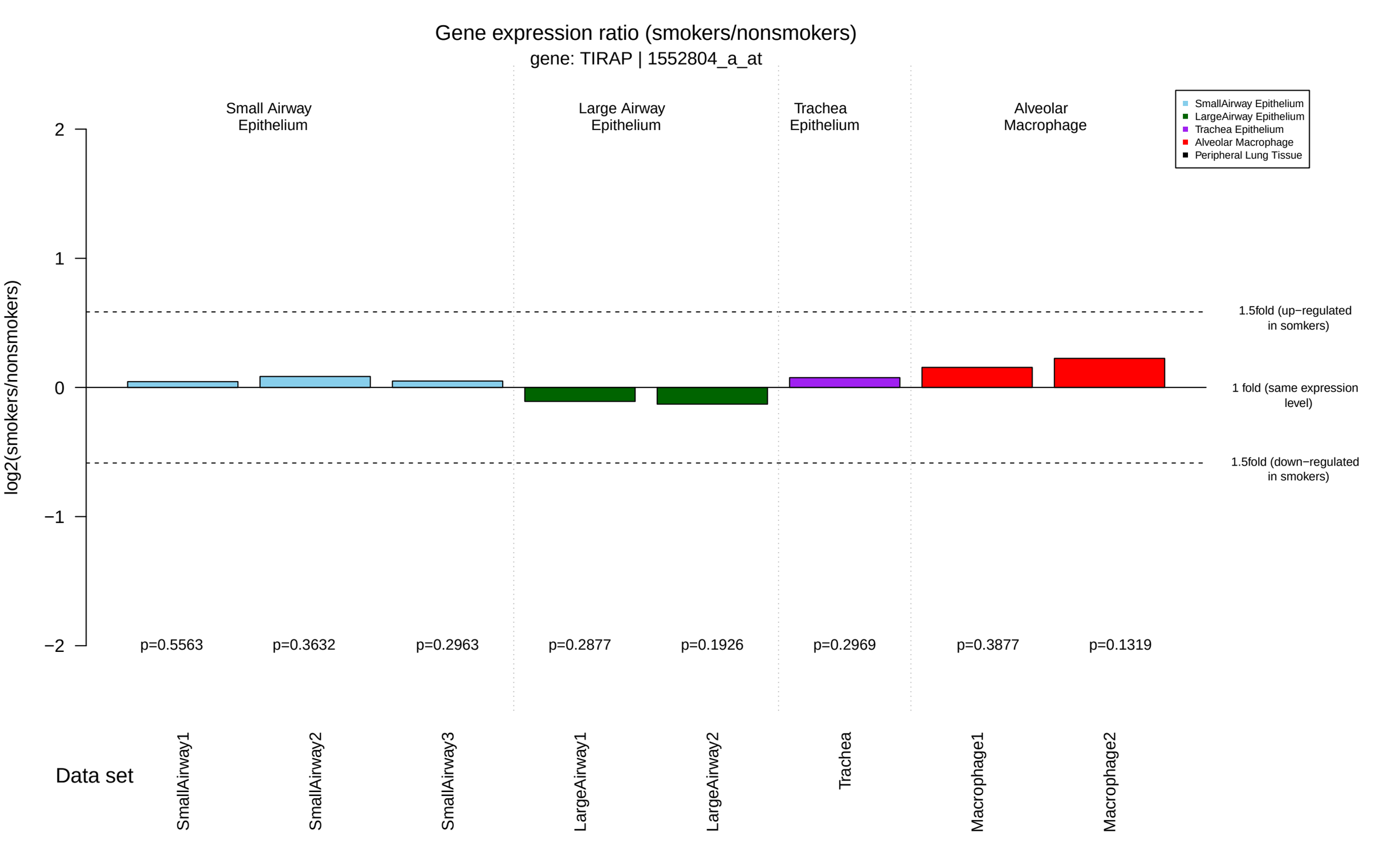
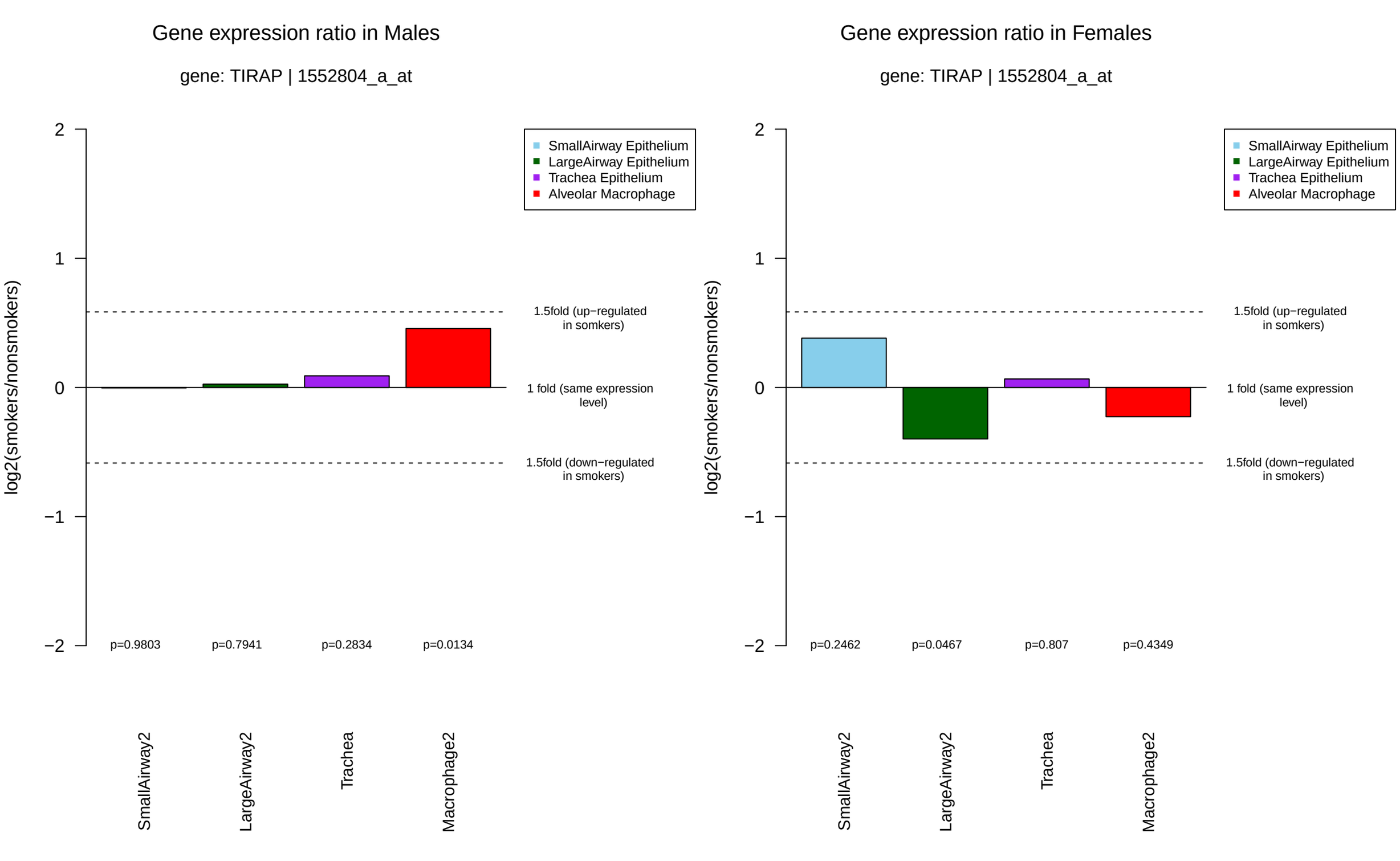
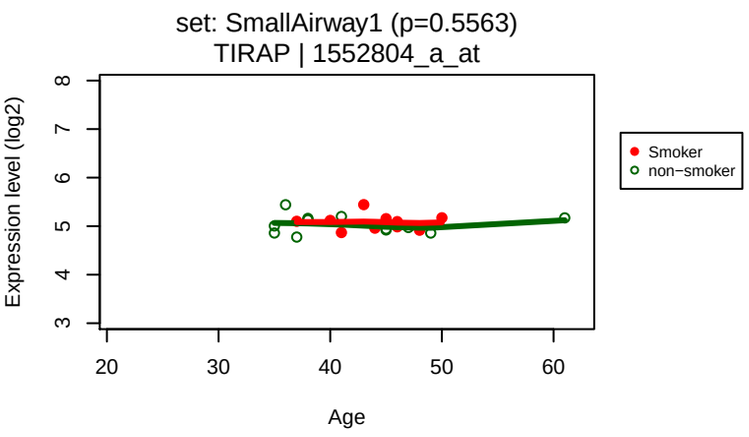
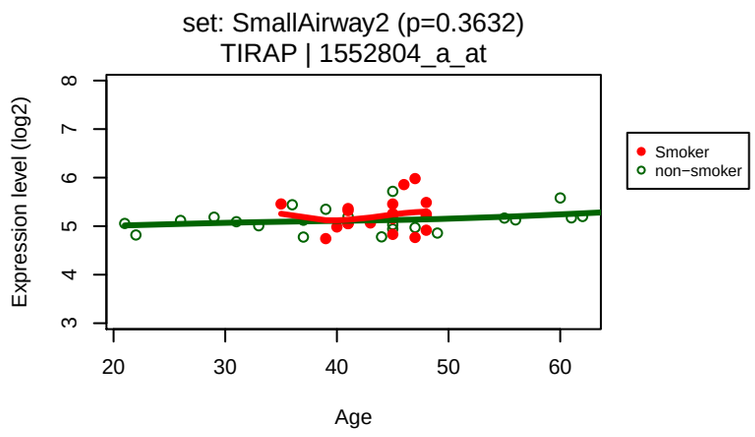
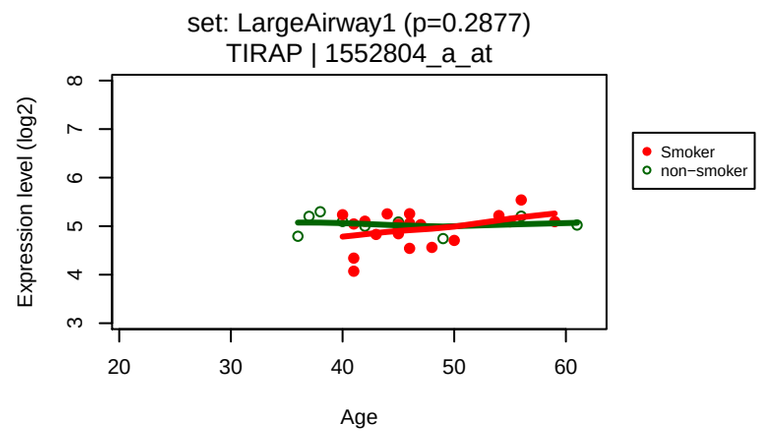

Gene expression of smokers and nonsmokers
gene: TIRAP | 1552804_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5563
p=0.3632
p=0.2963
p=0.2877
p=0.1926
p=0.2969
p=0.3877
p=0.1319
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

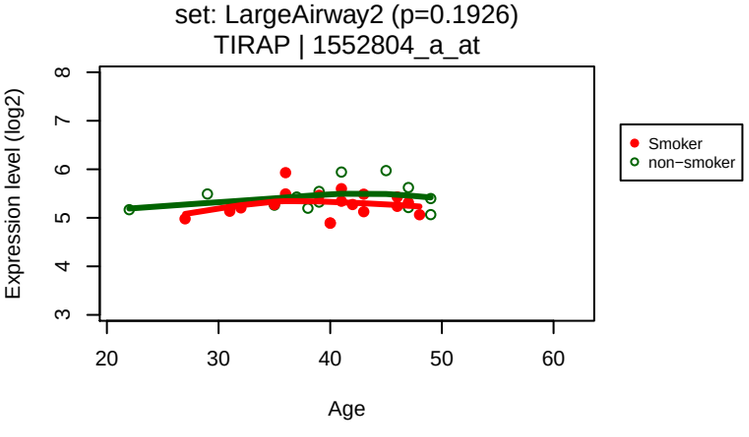

Gene expression ratio (smokers/nonsmokers)
gene: TIRAP | 1552804_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5563
p=0.3632
p=0.2963
p=0.2877
p=0.1926
p=0.2969
p=0.3877
p=0.1319
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

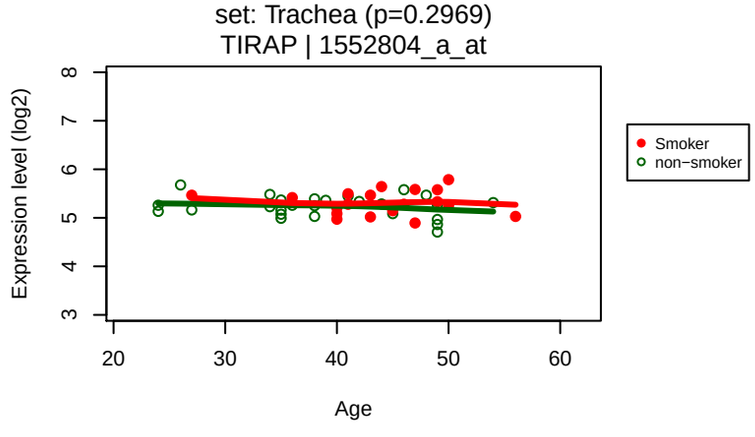

Gene expression ratio in Males
gene: TIRAP | 1552804_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9803
p=0.7941
p=0.2834
p=0.0134
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TIRAP | 1552804_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2462
p=0.0467
p=0.807
p=0.4349
SmallAirway2
LargeAirway2
Trachea
Macrophage2

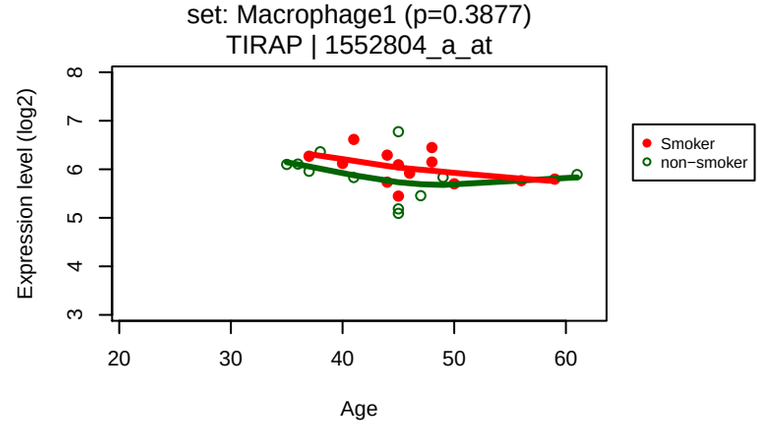

set: SmallAirway1 (p=0.5563)
TIRAP | 1552804_a_at
Expression level (log2)
Age
Smoker
non−smoker

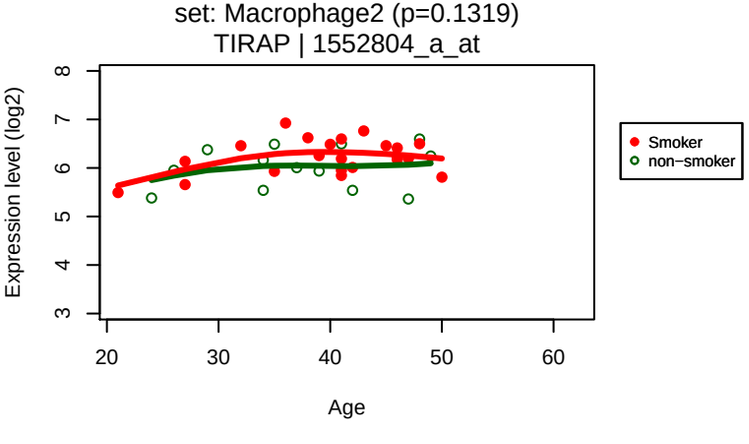

set: SmallAirway2 (p=0.3632)
TIRAP | 1552804_a_at
Expression level (log2)
Age
Smoker
non−smoker

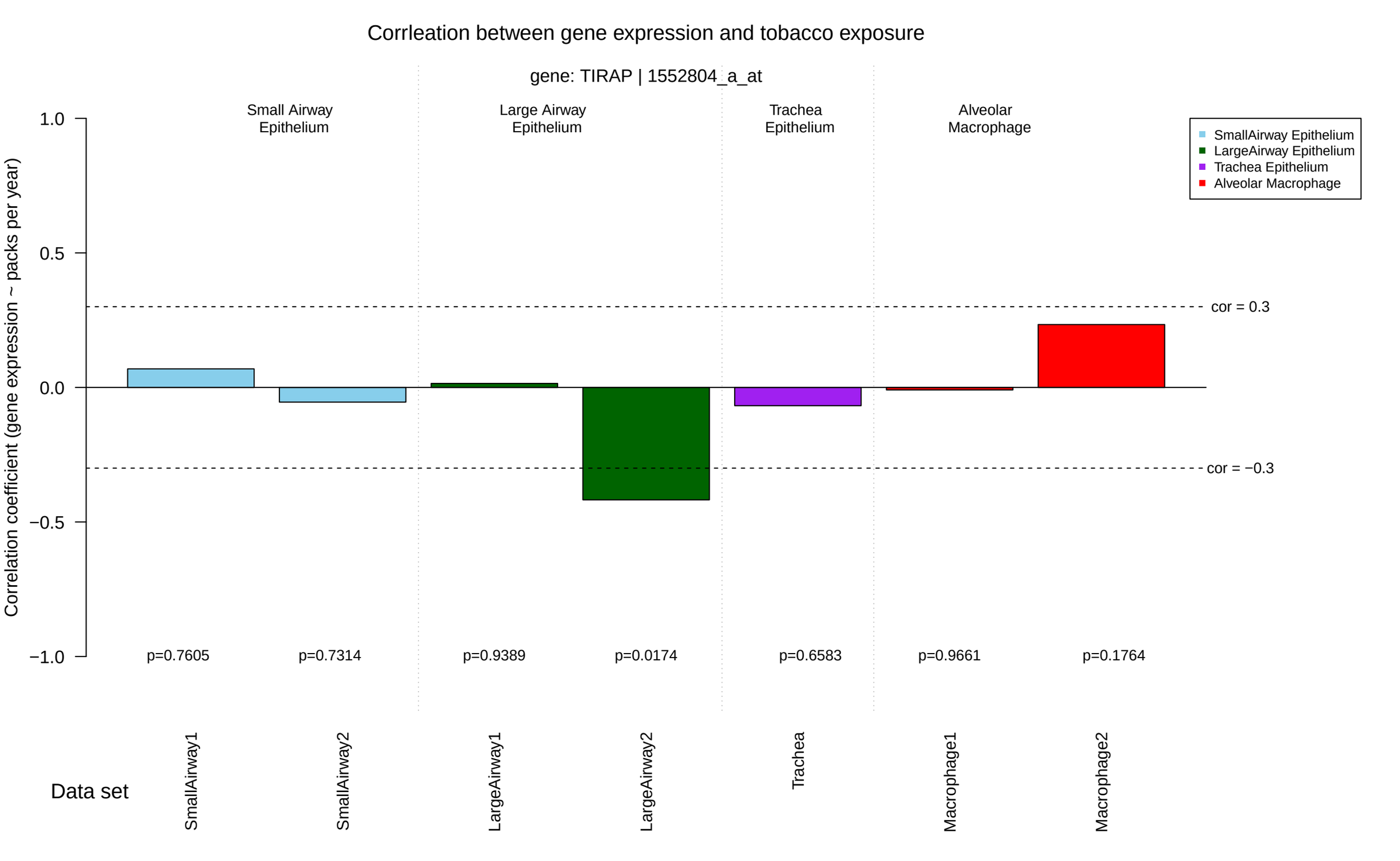

set: LargeAirway1 (p=0.2877)
TIRAP | 1552804_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1926)
TIRAP | 1552804_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2969)
TIRAP | 1552804_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3877)
TIRAP | 1552804_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1319)
TIRAP | 1552804_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TIRAP | 1552804_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7605
p=0.7314
p=0.9389
p=0.0174
p=0.6583
p=0.9661
p=0.1764
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2